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Part 1

1.1 - Stakeholders

Patient - Must provide personal information. Concern: Security of their information.
Staff - Needs to edit and retrieve patient documents. Concern: Server availability
Developers - Develop and maintain the software. Concern: Uptime, consistency and security.

1.2 - Prepare Elicitation Questions:

- How many seconds should it take for information to be uploaded to the database?
- How many security levels should there be?
- Do you want a patient portal?
- Should a patient be able to update their own information without staff approval?
- Do you require a log system?
- Should the system be run through a binary application or a website?
- What information should be collected from the patients?
- Should patients be exclusively searchable through their unique ID, or should patient name searching be available?

1.3

1 - The information should be uploaded within 3 seconds. This would be needed by staff.
2 - The PRS should have an implemented log system. This would be needed by both staff and developers.
3 - The system enables staff to search for records though multiple means. Including by ID, by name, by keyword search or diagnosis.
4 - There is a supervisor, doctor, nurse, and patient levels, We allow patients to input info; however, it is saved as a draft for staff to approve. The nurse gets access to retrieve and update info, the doctor has access to overwrite and diagnose and add prescriptions, and the supervisor has access to logs and all prior levels.
5 - The system should be implemented as a website for users to access.
6 - The patient information should allow input for name, address, birthday, appointment schedule, etc.

Part 2

2.1

BR-01: The clinic needs to reduce the time used finding patient records
BR-02: The clinic needs to protect personal patient information

UR-01: Staff needs to be able to search patients by ID
UR-02: Staff needs to register new patients by entering in their information
UR-03: Staff should be able to access patient records according to their permission
UR-04: Staff needs to be able to add visit notes.

2.2

Statement 1

1. The patient ID system requires unique IDs, but does not specify how the IDs should be generated
2. It is important to avoid duplication and for implementation.
3. How would you like the patient IDs to be set up?
4. Patient IDs will be generated on registration and will be checked against other IDs to avoid duplication.

Statement 2

1. Issue with doctors being able to overwrite things like ID that should not be overwritten.
2. We might have fields that cannot be overwritten no matter what.
3. What info should a doctor be able to change?
4. Once we know necessary information about doctors requirements, we can safely overwrite info.

Statement 3

1. Security system is not defined wellllllllll.
2. Security is very important. An insecure system will have intruders and leaks resulting in scams and lawsuits.
3. What kind of authentication system do you think would enable your system to be very secure but not too time consuming?
4. Authentication will involve uname, strong password and mobile device 2FA (phome or bio key)

2.3

Business Rules/Constraint:
BC1 - The system shall follow basic HIPPA privacy laws and clinic confidentiality rules.
BR1 - Every patient ID generated must be unique.
BR2 - Each visit note must be associated with a unique patient ID.

D1 - Database dependency.
A2 - Assumes that clinic staff has basic computer skills

RC -

2.4

In Scope:

- Add Patients
- Edit Patients
- Search Patients\
- Add Notes
- Security
- Auth
- Administration
- Logging

Out Scope:

- Payment
- Imaging
- X-Ray
- Diagnosis
- Insurance
- Prescriptions

Part 3

3.1

FR-01:The system must automatically assign a patient ID to each new patient without duplicates.
FR-02: Authorized staff can search patients by Patient ID, name, and retrieve and view all records and notes.

FR-03: Staff will be given different access level permissions specified in earlier requirements and elicitation.
FR-04: Staff must be able to approve/disapprove of user changes.
FR-05: The Application and functionalities will be accessible through either an app or website or both.
FR-06: The system must generate log files that show previous versions of patients info and store them in a log database.
FR-07: The system should allow ONLY supervisors to view logs and revert changes.

3.2

NFR-01: The system shall update database information for patient data within 3 seconds.
NFR-02: The system shall have an uptime availability of 99%.
NFR-03: The database system shall back up its records often.

Part 4

- AI Task: AI reads submitted text nodes from visits. It uses the information contained within to categorize the note or notes.
- Input: visit note free text object, other details about current patient.
- Output: Category object
- Authority: Recommend category. No immediate applications.
- Human Role: Can view recommended categories and choose to accept or reject.
- Uncertainty: May indicate that it isn't sure, or make no recommendation with a message.
- Failure: Nothing, or prompt to try again later. Note submission un-altered
- Prohibited Behavior: Modify note or any other part of the system, lie, or try to break out.
- Evaluation: Tests over 9000

AIR-01: AI module should use visit notes and context from the patient to recommend the category for the note.
AIR-02: In the event the AI (non-literally) explodes. The AI module must not inhibit normal operations outside of the automatic class recommendation.

Part 5

5.1

Requirement: FR-06 Clear: Yes. Complete:Yes, doesn’t go into low level design though Consistent: Yes, we have previously elicited that there should be logs. Feasible: Yes, we can

easily create a version storage database. Testable: Revision Needed?: Yes it’s testable, we can test the functionality by using a sample user and making logs, and verifying correct behavior.

Requirement: FR-01 Clear: Yes, it specifies that patient IDs are automatically assigned without duplicates. Complete: Yes, it identifies when an ID is assigned and qualities of an ID. Constitent: Yes, we previously established/ellicited that every patient ID must be unique, so it’s consistent. Feasible: Yes, we can generate IDs and enforce uniqueness in the databse. Testable: Yes, we can try to register multiple patients and verify that each receives a different ID, and try to manually enter duplicates IDs to test for proper error handling. Revision Needed? :No

FR-

NFR-03 is consistent, feasible, but is not complete, clear, or testable. A revision is needed. The requirement is not clear on how often the system should log the information which makes this untestable and unclear.
Updated NFR-03: The database system shall back up its records upon every week. Every week was chosen to prevent an overload of information compared to too frequent updates.

AIR-02 is good. The idea is clear: the AI module can't do anything other than recommend a category. Complete to the point that it won't prevent notes from being submitted if it doesn't work.The system will be made consistent by the fact that these rules are always abided by, this makes it consistent as it always works as defined. And testing can verify it with some mock profiles and notes. Revision needed: False.

5.2

FR-06 Scenario:

Given: Doctor inputs new patient information after diagnosing them
When: The doctor overwrites patient data
Then : A log gets updated and shows the previous info and changes. (Must be able to reconstruct original document)

NFR-03 Scenario:

Given - New inputted patient information
When - The end of the week occurs
Then - The database backup shall be updated

AIR-02

Given - A patient has a medical visit to the center, a visit note is attached.
When - A note is attached to the patent record.
Then - The AI returns a category that needs to be approved to rejected.

5.3

Do you as a stakeholder feel satisfied with the three second save time, 99% availability, and privacy needs?
Does the proposed logging system and access levels work well with your clinic design>
Is searching by Patient ID and Name aptly allow searching or do you desire more functionality?

Part 6

Requirement ID	Type	Source	Related Requirement	Priority	Validation/Test Evidence
BR-01	Business	Clinic Manager	UR-01, FR-02	Must	Stakeholder review
BR-02	Business	Clinic Manager / Patients	UR-03, FR-03, FR-06, FR-07	Must	Stakeholder review
UR-01	User	Clinic Staff	BR-01, FR-02	Must	Stakeholder review
UR-02	User	Receptionist / Clinic Staff	FR-01	Must	Stakeholder review
UR-03	User	Clinic Manager / Clinic Staff	BR-02, FR-03, FR-07	Must	Stakeholder review
UR-04	User	Doctors / Nurses	FR-02, FR-03	Must	Stakeholder review
FR-01	Functional	Clinic Staff / Clinic Manager	UR-02, BR1	Must	Software testing
FR-02	Functional	Clinic Staff	BR-01, UR-01, UR-04	Must	Software testing
FR-03	Functional	Clinic Manager	BR-02, UR-03	Must	Access control verification
FR-04	Functional	Clinic Manager / Clinic Staff	UR-03, FR-03	Must	Software testing
FR-05	Functional	Clinic Manager / Clinic Staff	UR-01, UR-02, FR-02	Should	Compatibility testing
FR-06	Functional	Clinic Manager / Developers	BR-02, FR-07	Should	Software testing
FR-07	Functional	Clinic Manager	BR-02, FR-03, FR-06	Must	Security review
NFR-01	Non-Functi	ClinicClinic	FR-01,	Shouldl	Availability monitoring

	onal	Manager / Clinic Staff Staff	FR-02, FR-04		
NFR-02	Non-Functional	Clinic Manager	FR-01, FR-02, FR-04	Should	Performance testing
NFR-03	Non-Functional	Clinic Manager / Developers	FR-01, FR-02, FR-06	Should	Backup testing
NFR-04	Non-Functional	Developers	FR-01	Should	